

Mathematical Methods · Harder paper

Real VCAA questions ranked by the published state results. Editable: type your answers straight into this PDF, save it, then mark yourself with the guide at the back.

Name _____ Date _____

Exam 1 ·

42 marks

reading 15 min · writing 60 min

Exam 2 ·

84 marks

reading 15 min · writing 120 min

State average on these questions

49 / 126 (39%)

a student at the state average on every question

How to use this paper

1. Open each question with its blue link. It opens the official VCAA exam PDF at the right page. The question text isn't reproduced here because it is VCAA copyright (some exams include third-party material).
2. Answer in this file: click a bubble for multiple choice, and type in the boxes for written answers (long answers scroll inside the box). Methods and science working can be typed in plain text, e.g. $v = \sqrt{2gh} = 9.9 \text{ m/s}$.
3. Keep to the real time limits above, and don't look at the guide until you've finished.
4. Save with Ctrl/Cmd + S. Use a PDF app that supports forms: Adobe Acrobat Reader, Chrome, Edge, Firefox, macOS Preview or Xodo.
5. Mark yourself: in the guide at the back, tick each marking point you earned, enter your marks, and compare your total with the state average.

Why these questions

Every question is a real VCAA exam question, chosen by ranking all the questions from the listed exams by how Victorian students actually scored on them (percentage correct for multiple choice, state average ÷ marks for written questions). This paper takes the hardest questions.

Sources

- [2023 Exam 1 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 1 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 1 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2023 Exam 2 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2024 Exam 2 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)
- [2025 Exam 2 Mathematical Methods exam \(PDF\)](#) · [examiners' report](#)

Exam 1 · technology free (42 marks, reading 15 min, writing 60 min)

Questions (technology free)

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (4 marks)

2023 VCAA Exam 1 · Question 6 · [open in the official exam PDF \(p. 7\)](#) ↗

Working backwards from an approximate 95% confidence interval (0.04, 0.16)

a. $\hat{p}$ used (1 mark)

b. Sample size ($z = 2$) (2 marks)

c. Effect on the CI width of a sample four times as big (1 mark)

Question 2 (6 marks)

2023 VCAA Exam 1 · Question 8 · [open in the official exam PDF \(p. 10\)](#) ↗

Queueing-time pdf $f(t) = kt(16 - t^2)$ on $[0, 4]$: find k , $E(T)$, and a conditional probability

a. Show that $k = \frac{1}{64}$ (1 mark)

b. $E(T)$ (2 marks)

c. $\Pr(T > 2 \mid T > 1)$ (3 marks)

Question 3 (4 marks)

2024 VCAA Exam 1 · Question 6 · [open in the official exam PDF \(p. 1\)](#) ↗

Solve $2 \log_3(x - 4) + \log_3(x) = 2$

Solve for x (4 marks)

Question 4 (9 marks)

2024 VCAA Exam 1 · Question 7 · [open in the official exam PDF \(p. 8\)](#) ↗

$f(x) = x \sin(x)$: trapezium approximation, the range of f' on an interval, a stationary point, sketching f'

a. Trapezium rule, step $\frac{\pi}{3}$, on $[0, \pi]$ (3 marks)

b.i. Find $f'(x)$ (1 mark)

b.ii. Range of f' over $\left[\frac{\pi}{2}, \frac{2\pi}{3}\right]$ (1 mark)

b.iii. Hence verify a stationary point in that interval (1 mark)

c. Sketch $y = f'(x)$ on $[-\pi, \pi]$ with endpoints (3 marks)

Question 5 (7 marks)

2024 VCAA Exam 1 · Question 8 · [open in the official exam PDF \(p. 10\)](#) ↗

Cube-root function $g(x) = \sqrt[3]{x-k} + m$: y-intercept, gradient there, through the origin, a second point with equal gradient

a. Coordinates of the y-intercept P (1 mark)

b. Gradient of g at P in terms of k (2 marks)

c. k in terms of m if the graph passes through the origin (1 mark)

d. Coordinates of Q ($\neq$ P) with the same gradient, graph through the origin (3 marks)

Question 6 (5 marks)

2025 VCAA Exam 1 · Question 8 · [open in the official exam PDF \(p. 10\)](#) ↗

Continuous pdf: find k with $\Pr(X > k) = 9/16$; integral of a transformed pdf

a. Find k such that $\Pr(X > k) = \frac{9}{16}$ (3 marks)

b. $\int_0^{4/3} h(x) dx$ for $h(x) = mf(x) + n$ (2 marks)

Question 7 (7 marks)

2025 VCAA Exam 1 · Question 9 · [open in the official exam PDF \(p. 12\)](#) ↗

Solving $\frac{w^2}{(x-1)^2} = (x-w)^2$; the minimum of $(x-1)(x-w)$; when there are exactly three solutions

a. Four solutions when $w = -3$ (3 marks)

b.i. Minimum point of $y = (x - 1)(x - w)$, $w > 0$ (2 marks)

b.ii. Positive w giving exactly three solutions (2 marks)

Exam 2 · CAS and a bound reference allowed (84 marks, reading 15 min, writing 120 min)

Section A – Multiple-choice questions

Open each question in the official exam PDF (click the blue link), then select A–E. One mark each.

Question 1 · 1 mark

2023 VCAA Exam 2 · Section A · Question 9 · [open in the official exam PDF \(p. 5\)](#) ↗

Value of a that makes a $\tan/\sin$ hybrid continuous and smooth at $x = 2\pi$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 2 · 1 mark

2023 VCAA Exam 2 · Section A · Question 11 · [open in the official exam PDF \(p. 5\)](#) ↗

Gradient of $f \times g$ at $x = -2$ from given values

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 3 · 1 mark

2023 VCAA Exam 2 · Section A · Question 12 · [open in the official exam PDF \(p. 6\)](#) ↗

Maximum possible mean of a pmf with parameter k

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 4 · 1 mark

2023 VCAA Exam 2 · Section A · Question 14 · [open in the official exam PDF \(p. 7\)](#) ↗

Number of tangents to a cubic that pass through its positive x -intercept

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 5 · 1 mark

2023 VCAA Exam 2 · Section A · Question 17 · [open in the official exam PDF \(p. 8\)](#) ↗

Volume of a cylinder cut from a sheet, in terms of x and y

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 6 · 1 mark

2023 VCAA Exam 2 · Section A · Question 18 · [open in the official exam PDF \(p. 9\)](#) ↗

Number of local minima of $\sin(ax)$ on $[-a\pi, a\pi]$

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 7 · 1 mark

2023 VCAA Exam 2 · Section A · Question 19 · [open in the official exam PDF \(p. 9\)](#) ↗

Values of k giving one positive and one negative root of a quadratic

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 8 · 1 mark

2023 VCAA Exam 2 · Section A · Question 20 · [open in the official exam PDF \(p. 9\)](#) ↗

Largest interval where both $f \circ g$ and $g \circ f$ exist (log and sine)

Your answer: ☐ A ☐ B ☐ C ☐ D ☐ E

Question 9 · 1 mark2024 VCAA Exam 2 · Section A · Question 10 · [open in the official exam PDF \(p. 4\)](#) ↗What must be true if $f'(2) < 0$ and $f'(4) > 0$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 10 · 1 mark2024 VCAA Exam 2 · Section A · Question 11 · [open in the official exam PDF \(p. 5\)](#) ↗

Choosing 3 of 12 students: probability exactly two are from the front row of seven

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 11 · 1 mark2024 VCAA Exam 2 · Section A · Question 13 · [open in the official exam PDF \(p. 7\)](#) ↗Local minimum of $f(x) = \frac{x}{2} + \frac{2}{x}$ after dilation $\times 3$ from the y-axis and a shift down 1Your answer: ☐ A ☐ B ☐ C ☐ D

Question 12 · 1 mark2024 VCAA Exam 2 · Section A · Question 16 · [open in the official exam PDF \(p. 8\)](#) ↗Gradient of $y = \sqrt{f(x)}$ at $x = 4$ given $f(4) = 25$ and $f'(4) = 15$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 13 · 1 mark2024 VCAA Exam 2 · Section A · Question 17 · [open in the official exam PDF \(p. 9\)](#) ↗

Output of an algorithm that searches for integer roots of a cubic

Your answer: ☐ A ☐ B ☐ C ☐ D

Question 14 · 1 mark2024 VCAA Exam 2 · Section A · Question 20 · [open in the official exam PDF \(p. 10\)](#) ↗ $\int_2^6 f(x) dx$ for a period-2 function with average value k on $[0, 2]$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 15 · 1 mark2025 VCAA Exam 2 · Section A · Question 13 · [open in the official exam PDF \(p. 8\)](#) ↗Graph of $g(f(x))$ given sketches of a line f and an even function g Your answer: ☐ A ☐ B ☐ C ☐ D

Question 16 · 1 mark2025 VCAA Exam 2 · Section A · Question 15 · [open in the official exam PDF \(p. 9\)](#) ↗Image of the point $(1, 3)$ on $y = 1 - g(2x - 3)$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 17 · 1 mark2025 VCAA Exam 2 · Section A · Question 16 · [open in the official exam PDF \(p. 9\)](#) ↗Conditions on a, b if $h(x) = a \log_e(bx)$ has $h'(x) > 0$ everywhereYour answer: ☐ A ☐ B ☐ C ☐ D

Question 18 · 1 mark2025 VCAA Exam 2 · Section A · Question 17 · [open in the official exam PDF \(p. 10\)](#) ↗Which graph fits $\int_1^2 f dx > \int_1^3 f dx$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 19 · 1 mark2025 VCAA Exam 2 · Section A · Question 19 · [open in the official exam PDF \(p. 12\)](#) ↗Value of c when the shortest distance from $y = x + c$ to $y = \log_e(x - 1)$ is $\sqrt{2}$ Your answer: ☐ A ☐ B ☐ C ☐ D

Question 20 · 1 mark2025 VCAA Exam 2 · Section A · Question 20 · [open in the official exam PDF \(p. 12\)](#) ↗Which transformation sequence does not map a^x to a^{2x+2} Your answer: ☐ A ☐ B ☐ C ☐ D

Section B – Extended-response questions

Open each question in the official exam PDF, then type your answer to every part in the boxes. Show working where more than one mark is available.

Question 1 (11 marks)2023 VCAA Exam 2 · Section B · Question 2 · [open in the official exam PDF \(p. 12\)](#) ↗Observation wheel modelled by $h(t) = -60\cos(bt) + c$; then it stops and restarts at double speed**a.** Show $b = \frac{\pi}{15}$ and $c = 75$ (2 marks)**b.** Average height from A (bottom) to B (level with the centre) (2 d.p.) (2 marks)**c.** Average rate of change of height from A to B (1 mark)**d.i.** k and m in the piecewise model (1 mark)

d.ii. All possible values of n (2 marks)

d.iii. Sketch w with the coordinates of the endpoints (3 marks)

Question 2 (12 marks)

2023 VCAA Exam 2 · Section B · Question 3 · [open in the official exam PDF \(p. 16\)](#) ↗

$g(x) = 2^x + 5$: limit, derivative, tangents through the origin; $h(x) = 2^x - x^2$: inflection, decreasing interval, Newton's method

a. $\lim_{x \rightarrow -\infty} g(x)$ (1 mark)

b. k with $g'(x) = k \times 2^x$ (1 mark)

c.i. Tangent at $(a, g(a))$ (1 mark)

c.ii. Tangent to g through the origin (3 d.p.) (2 marks)

d. Point of inflection of h (2 d.p.) (1 mark)

e. Largest interval on which h is strictly decreasing (2 d.p.) (1 mark)

f. Newton's method from $x_0 = 0$: x_1, x_2, x_3 (3 d.p.) (2 marks)

g. Why a solution of $\log_e(2)2^x - 2x = 0$ is a bad starting point (1 mark)

h. Positive n for which $n^x - x^n$ has a local minimum on the x -axis (2 marks)

Question 3 (11 marks)

2023 VCAA Exam 2 · Section B · Question 5 · [open in the official exam PDF \(p. 21\)](#) ↗

$g(x) = \frac{1}{2}f(2-x)$ with $f(x) = e^x + e^{-x}$: transformations, restricted inverses, areas with inverses, a family of curves

a. Complete the sequence of transformations from f to g (2 marks)

b. Domain and range of the inverse of the increasing branch g_1 (2 marks)

c.i. Coordinates of P and Q (2 d.p.) (1 mark)

c.ii. Area bounded by g and the two inverses (2 d.p.) (2 marks)

d. Integer n such that the turning point of h always lies on $y = 2x^n$ (1 mark)

e. Smallest k for which h meets the inverse of h_1 (2 d.p.) (1 mark)

f. Area between h and the inverse of h_1 when $k = 5$ (2 d.p.) (2 marks)

Question 4 (11 marks)

2024 VCAA Exam 2 · Section B · Question 5 · [open in the official exam PDF \(p. 24\)](#) ↗

Composites of $\sin x$ and $\sin 2x$: maxima, ranges, derivatives, stationary points, area between them, domain of a composite

a.i. Local maximum of $g \circ f$ in $[0, \frac{\pi}{2}]$ (1 d.p.) (1 mark)

a.ii. Range of $g \circ f$ on $[0, 2\pi]$ (1 mark)

b.i. Derivative of $f \circ g$ (1 mark)

b.ii. Show $\cos(\sin 2x) = 0$ has no real solutions (2 marks)

b.iii. x-values of the stationary points of $f \circ g$ on $[0, 2\pi]$ (1 mark)

b.iv. Range of $f \circ g$ (1 mark)

c.i. Single definite integral for the area between the two composites on $[0, 2\pi]$ (1 mark)

c.ii. That area (2 d.p.) (1 mark)

d. x in $(0, 2\pi)$ for which $f_1 \circ g$ is defined, f_1 having domain $(0, 2\pi)$ (2 marks)

Question 5 (19 marks)

2025 VCAA Exam 2 · Section B · Question 4 · [open in the official exam PDF \(p. 23\)](#) ↗

$f(x) = \sin x + 1$ on $[0, \frac{5\pi}{2}]$: values, periodicity, tangents, Newton's method, a family of tangents, matching a polynomial

a. $f(\frac{2\pi}{3})$ (1 mark)

b. Exact x with $f(x) = \frac{3}{2}$ (1 mark)

c. k and the largest a with $f(x+k) = f(x)$ on $[0, a]$ (2 marks)

d. Tangent at $x = \frac{2\pi}{3}$ (1 mark)

e.i. Newton's method from $x_0 = \frac{2\pi}{3}$: x_2 to 1 d.p. (1 mark)

e.ii. Draw the tangent at $x = x_1$ on the graph (1 mark)

f.i. Show the tangent at $x = p$ is $t(x) = \cos(p)(x - p) + \sin(p) + 1$ (2 marks)

f.ii. Minimum and maximum y-intercepts of t for $p \in (0, \frac{5\pi}{2})$ (2 marks)

f.iii. p for which t has a unique x-intercept equal to f 's x-intercept (2 d.p.) (2 marks)

g.i. Show $c = 1$ and $d = 1$ given $g(0) = f(0)$, $g'(0) = f'(0)$ (2 marks)

g.ii. Area between f and g on $[0, 2\pi]$ when they also match at 2π (2 d.p.) (2 marks)

g.iii. b and r with $g(r) = f(r)$, $g'(r) = f'(r)$ when $a = 0$, $c = d = 1$ (2 marks)

END OF PAPER

Marking guide follows

Stop here until you have finished the paper under exam conditions.

Marking guide · Exam 1

Questions (technology free)

Question 1 (4 marks) · state average 1.4 / 4 (35%)

2023 VCAA Exam 1 · Question 6

a. $\hat{p}$ used

☐ 0.1

Answer / notes: The midpoint: $\hat{p} = 0.10$.

State average 0.5 / 1 (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: _____ / 1

b. Sample size ($z = 2$)

☐ Margin $0.06 = 2\sqrt{\frac{0.1 \times 0.9}{n}}$

☐ $n = 100$

Answer / notes: $\sqrt{\frac{0.09}{n}} = 0.03 \Rightarrow n = 100$

State average 0.7 / 2 (35%); 26% scored full marks. Mark distribution: 0: 51% · 1: 23% · 2: 26%

Your mark: _____ / 2

c. Effect on the CI width of a sample four times as big

☐ Width is halved (factor $\frac{1}{2}$)

Answer / notes: Width $\propto \frac{1}{\sqrt{n}}$, so $\times \frac{1}{2}$.

State average 0.2 / 1 (20%); 23% scored full marks. Mark distribution: 0: 77% · 1: 23%

Your mark: _____ / 1

Question 2 (6 marks) · state average 1.9 / 6 (32%)

2023 VCAA Exam 1 · Question 8

a. Show that $k = \frac{1}{64}$

☐ $k \int_0^4 (16t - t^3) dt = 64k = 1$

Answer / notes: $k \left[8t^2 - \frac{t^4}{4} \right]_0^4 = 64k = 1$

State average 0.4 / 1 (40%); 43% scored full marks. Mark distribution: 0: 57% · 1: 43%

Your mark: _____ / 1

b. $E(T)$

☐ $\frac{1}{64} \int_0^4 (16t^2 - t^4) dt$

☐ $= \frac{32}{15}$

Answer / notes: $\frac{1}{64} \left(\frac{1024}{3} - \frac{1024}{5} \right) = \frac{32}{15}$

State average 0.8 / 2 (40%); 26% scored full marks. Mark distribution: 0: 50% · 1: 24% · 2: 26%

Your mark: _____ / 2

c. $\Pr(T > 2 \mid T > 1)$

☐ $\Pr(T > 2) = \frac{9}{16}$

☐ $\Pr(T > 1) = \frac{225}{256}$

☐ Ratio $= \frac{16}{25}$

Answer / notes: $\frac{9/16}{225/256} = \frac{16}{25}$

State average 0.7 / 3 (23%); 11% scored full marks. Mark distribution: 0: 63% · 1: 13% · 2: 13% · 3: 11%

Your mark: _____ / 3

Question 3 (4 marks) · state average 1.5 / 4 (38%)

2024 VCAA Exam 1 · Question 6

Solve for x

- ☐ Combines: $\log_3 (x(x-4)^2) = 2$
- ☐ $x^3 - 8x^2 + 16x - 9 = 0$
- ☐ Factorises: $(x-1)(x^2 - 7x + 9) = 0$
- ☐ Only $x = \frac{7+\sqrt{13}}{2}$ satisfies $x > 4$

Answer / notes: $x = \frac{7+\sqrt{13}}{2}$. Reject $x = 1$ and $\frac{7-\sqrt{13}}{2}$ because the log needs $x > 4$.

State average 1.5 / 4 (38%); 9% scored full marks. Mark distribution: 0: 18% · 1: 52% · 2: 8% · 3: 13% · 4: 9%

Your mark: _____ / 4

Question 4 (9 marks) · state average 3.8 / 9 (42%)

2024 VCAA Exam 1 · Question 7

a. Trapezium rule, step $\frac{\pi}{3}$, on $[0, \pi]$

- ☐ Values: $f(0) = 0$, $f(\frac{\pi}{3}) = \frac{\sqrt{3}\pi}{6}$, $f(\frac{2\pi}{3}) = \frac{\sqrt{3}\pi}{3}$, $f(\pi) = 0$
- ☐ Correct trapezium formula applied
- ☐ $\frac{\sqrt{3}\pi^2}{6}$

Answer / notes: $\frac{\pi}{6} \left[0 + 2 \left(\frac{\sqrt{3}\pi}{6} + \frac{\sqrt{3}\pi}{3} \right) + 0 \right] = \frac{\sqrt{3}\pi^2}{6}$

State average 1.5 / 3 (50%); 30% scored full marks. Mark distribution: 0: 28% · 1: 31% · 2: 11% · 3: 30%

Your mark: _____ / 3

b.i. Find $f'(x)$

- ☐ $\sin(x) + x \cos(x)$

Answer / notes: $f'(x) = \sin(x) + x \cos(x)$

State average 0.9 / 1 (90%); 84% scored full marks. Mark distribution: 0: 16% · 1: 84%

Your mark: _____ / 1

b.ii. Range of f' over $[\frac{\pi}{2}, \frac{2\pi}{3}]$

- ☐ $[\frac{\sqrt{3}}{2} - \frac{\pi}{3}, 1]$

Answer / notes: $f'(\frac{\pi}{2}) = 1$ and $f'(\frac{2\pi}{3}) = \frac{\sqrt{3}}{2} - \frac{\pi}{3}$ (decreasing between them).

State average 0.2 / 1 (20%); 20% scored full marks. Mark distribution: 0: 80% · 1: 20%

Your mark: _____ / 1

b.iii. Hence verify a stationary point in that interval

- ☐ f' is continuous and changes sign ($1 > 0$ and $\frac{\sqrt{3}}{2} - \frac{\pi}{3} < 0$), so $f'(x) = 0$ somewhere in the interval

Answer / notes: The range contains 0 and f' is continuous, so it must hit zero. Only 12% got this.

State average 0.1 / 1 (10%); 12% scored full marks. Mark distribution: 0: 88% · 1: 12%

Your mark: _____ / 1

c. Sketch $y = f'(x)$ on $[-\pi, \pi]$ with endpoints

- ☐ Endpoints $(-\pi, \pi)$ and $(\pi, -\pi)$
- ☐ Odd-function shape through the origin with the given min $(-1.1, -1.4)$ and max $(1.1, 1.4)$
- ☐ x-intercepts where f has turning points ($\approx \pm 2.03$)

Answer / notes: An odd curve from $(-\pi, \pi)$ down through ≈ -2 to the minimum, up through O to the maximum, then down through ≈ 2 to $(\pi, -\pi)$.

State average 1.1 / 3 (37%); 15% scored full marks. Mark distribution: 0: 45% · 1: 16% · 2: 24% · 3: 15%

Your mark: _____ / 3

Question 5 (7 marks) · state average 2.4 / 7 (34%)

2024 VCAA Exam 1 · Question 8

a. Coordinates of the y-intercept P

☐ $(0, m - \sqrt[3]{k})$

Answer / notes: $P = (0, m - \sqrt[3]{k})$ (the cube root of $-k$ is $-\sqrt[3]{k}$).

State average 0.6 / 1 (60%); 61% scored full marks. Mark distribution: 0: 39% · 1: 61%

Your mark: _____ / 1

b. Gradient of g at P in terms of k

☐ $g'(x) = \frac{1}{3}(x - k)^{-2/3}$

☐ $g'(0) = \frac{1}{3k^{2/3}}$

Answer / notes: $\frac{1}{3\sqrt[3]{k^2}}$. Keep it positive: $(-k)^{2/3} = k^{2/3}$.

State average 0.8 / 2 (40%); 32% scored full marks. Mark distribution: 0: 55% · 1: 13% · 2: 32%

Your mark: _____ / 2

c. k in terms of m if the graph passes through the origin

☐ $k = m^3$

Answer / notes: $m - \sqrt[3]{k} = 0 \Rightarrow k = m^3$

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: _____ / 1

d. Coordinates of Q ($\neq$ P) with the same gradient, graph through the origin

☐ $(x - k)^{2/3} = k^{2/3} \Rightarrow x - k = \pm k$

☐ $x = 2k$ ($x = 0$ is P)

☐ $Q = (2m^3, 2m)$

Answer / notes: By the symmetry of the cube root about its inflection point (k, m) : $Q = (2m^3, 2m)$. The state averaged 0.5/3.

State average 0.5 / 3 (17%); 9% scored full marks. Mark distribution: 0: 74% · 1: 15% · 2: 2% · 3: 9%

Your mark: _____ / 3

Question 6 (5 marks) · state average 2.2 / 5 (44%)

2025 VCAA Exam 1 · Question 8

a. Find k such that $\Pr(X > k) = \frac{9}{16}$

☐ Sets up $\int_k^{4/3} \frac{3}{8}(4 - 3x) dx = \frac{9}{16}$ (or the complement)

☐ Gets $9k^2 - 24k + 7 = 0$

☐ $k = \frac{1}{3}$ (rejects $\frac{7}{3}$, outside the domain)

Answer / notes: $k = \frac{1}{3}$. The other root $\frac{7}{3}$ is outside $[0, \frac{4}{3}]$.

State average 1.4 / 3 (47%); 24% scored full marks. Mark distribution: 0: 30% · 1: 23% · 2: 23% · 3: 24%

Your mark: _____ / 3

b. $\int_0^{4/3} h(x) dx$ for $h(x) = mf(x) + n$

☐ Uses $\int_0^{4/3} f(x) dx = 1$

☐ $m + \frac{4n}{3}$

Answer / notes: $m \cdot 1 + n \cdot \frac{4}{3} = m + \frac{4n}{3}$

State average 0.8 / 2 (40%); 27% scored full marks. Mark distribution: 0: 51% · 1: 22% · 2: 27%

Your mark: _____ / 2

Question 7 (7 marks) · state average 1.9 / 7 (27%)

2025 VCAA Exam 1 · Question 9

a. Four solutions when $w = -3$

☐ Reduces to $(x + 3)(x - 1) = \pm 3$

☐ $+3: x^2 + 2x - 6 = 0 \Rightarrow x = -1 \pm \sqrt{7}$

☐ $-3: x^2 + 2x = 0 \Rightarrow x = 0, -2$

Answer / notes: $x = -2, 0, -1 \pm \sqrt{7}$

State average 1.0 / 3 (33%); 19% scored full marks. Mark distribution: 0: 47% · 1: 25% · 2: 10% · 3: 19%

Your mark: _____ / 3

b.i. Minimum point of $y = (x - 1)(x - w)$, $w > 0$

☐ $x = \frac{w+1}{2}$

☐ $y = -\frac{(w-1)^2}{4}$

Answer / notes: $\left(\frac{w+1}{2}, -\frac{(w-1)^2}{4}\right)$

State average 0.8 / 2 (40%); 22% scored full marks. Mark distribution: 0: 43% · 1: 34% · 2: 22%

Your mark: _____ / 2

b.ii. Positive w giving exactly three solutions

☐ $(x - 1)(x - w) = w$ always gives 2 solutions, so $(x - 1)(x - w) = -w$ must touch the minimum: $-w = -\frac{(w-1)^2}{4}$

☐ $w^2 - 6w + 1 = 0 \Rightarrow w = 3 \pm 2\sqrt{2}$

Answer / notes: $w = 3 \pm 2\sqrt{2}$. The state averaged 0.1/2.

State average 0.1 / 2 (5%); 4% scored full marks. Mark distribution: 0: 94% · 1: 3% · 2: 4%

Your mark: _____ / 2

Marking guide · Exam 2

Section A – Multiple-choice questions

✓	Q	Ans	Why	How Victoria went
☐	1	C	Continuity needs $\sin(2\pi a) = 0$. Smoothness needs $a \cos(2\pi a) = \frac{1}{2}$ (the gradient of $\tan \frac{\pi}{2}$ there). So $a = -\frac{1}{2}$. <i>Most popular wrong answer: D (31%).</i>	42% correct A 7% · B 9% · C 42% · D 31% · E 10%
☐	2	E	Product rule: $3 \times 8 + (-7) \times 2 = 10$. <i>Most popular wrong answer: D (51%).</i>	22% correct A 6% · B 13% · C 8% · D 51% · E 22%
☐	3	E	All probabilities must lie in $[0, 1]$, so $0 \leq k \leq \sqrt{5} - 2$. The mean $-3k^2 - 7k + 2$ is largest at $k = 0$, giving 2. <i>Most popular wrong answer: C (26%).</i>	29% correct A 11% · B 14% · C 26% · D 18% · E 29%
☐	4	D	Set up the tangent at a general point and require it to pass through $(\frac{1}{3}, 0)$. The resulting cubic in the tangent point has 3 real solutions. <i>Most popular wrong answer: B (33%).</i>	29% correct A 6% · B 33% · C 22% · D 29% · E 10%
☐	5	B	The circumference $y = 2\pi r$ and $h = x - 4r$, so $V = \pi r^2 h = \frac{\pi x y^2 - 2y^3}{4\pi^2}$. <i>Most popular wrong answer: C (26%).</i>	28% correct A 6% · B 28% · C 26% · D 26% · E 13%
☐	6	E	Interval length $2a\pi \div \text{period } \frac{2\pi}{a} = a^2$ periods, each with one minimum. <i>Most popular wrong answer: D (26%).</i>	29% correct A 12% · B 9% · C 22% · D 26% · E 29%
☐	7	D	Opposite signs need the product of the roots to be negative: $4k^2 - \frac{9}{4} < 0$, so $-\frac{3}{4} < k < \frac{3}{4}$. <i>Most popular wrong answer: A (27%).</i>	32% correct A 27% · B 11% · C 13% · D 32% · E 16%
☐	8	A	$g \circ f$ needs $x > -\frac{1}{\sqrt{2}}$. $f \circ g$ needs $\sin x > -\frac{1}{\sqrt{2}}$, i.e. $x < \frac{5\pi}{4}$ here. Combined: $(-\frac{1}{\sqrt{2}}, \frac{5\pi}{4})$. <i>Most popular wrong answer: C (26%).</i>	30% correct A 30% · B 19% · C 26% · D 14% · E 9%
☐	9	B	f' is continuous and changes sign, so f is not one-to-one on $[0, 5]$ and has no inverse. The others might not hold. <i>Most popular wrong answer: A (26%).</i>	38% correct A 26% · B 38% · C 16% · D 20%
☐	10	B	$\frac{\binom{7}{2}\binom{5}{1}}{\binom{12}{3}} = \frac{105}{220} = \frac{21}{44}$ <i>Most popular wrong answer: D (37%).</i>	40% correct A 12% · B 40% · C 10% · D 37%
☐	11	A	f has its minimum at $(2, 2)$. The dilation gives $(6, 2)$, and the shift gives $(6, 1)$. <i>Most popular wrong answer: B (24%).</i>	45% correct A 45% · B 24% · C 17% · D 14%
☐	12	D	$\frac{f'}{2\sqrt{f}} = \frac{15}{10} = \frac{3}{2}$. <i>Most popular wrong answer: C (27%).</i>	36% correct A 22% · B 14% · C 27% · D 36%
☐	13	D	The roots are 3, -3 and 2. The loop counts c down from 18, checking c then $-c$, so it prints 3, -3 , 2. <i>Most popular wrong answer: C (32%).</i>	27% correct A 14% · B 26% · C 32% · D 27%
☐	14	C	Each period contributes $2k$, and $[2, 6]$ is two periods, so $4k$. <i>Most popular wrong answer: B (23%).</i>	41% correct A 20% · B 23% · C 41% · D 14%
☐	15	C	f is $y = -x$ and g is even, so $g(-x) = g(x)$. The composite looks just like g . <i>Most popular wrong answer: D (25%).</i>	45% correct A 12% · B 17% · C 45% · D 25%

✓	Q	Ans	Why	How Victoria went
☐	16	B	You need $2x - 3 = 1$, so $x = 2$ and $y = 1 - 3 = -2$. <i>Most popular wrong answer: A (25%).</i>	44% correct A 25% · B 44% · C 22% · D 8%
☐	17	D	$h'(x) = \frac{a}{x}$, with domain $bx > 0$. If $b > 0$ then $x > 0$ and you need $a > 0$. If $b < 0$ then $x < 0$ and you need $a < 0$. Either way $ab > 0$. <i>Most popular wrong answer: C (46%).</i>	18% correct A 20% · B 16% · C 46% · D 18%
☐	18	A	The condition means $\int_2^3 f(x) dx < 0$, so f must be mostly below the axis on $[2, 3]$. <i>Most popular wrong answer: D (24%).</i>	38% correct A 38% · B 20% · C 17% · D 24%
☐	19	D	The closest point on the curve has gradient 1: $(2, 0)$. Distance $\frac{ 2+c }{\sqrt{2}} = \sqrt{2}$ gives $c = 0$ or -4 , but $c = -4$ cuts the curve (distance 0), so $c = 0$. <i>Most popular wrong answer: A (35%).</i>	14% correct A 35% · B 25% · C 24% · D 14%
☐	20	C	Apply each sequence in order. The sequence “dilate by a from the x-axis, dilate by $\frac{1}{2}$ from the y-axis, then translate 1 right” gives a^{2x-1} . <i>Most popular wrong answer: A (24%).</i>	36% correct A 24% · B 18% · C 36% · D 21%

Multiple-choice score: _____ / 20

Section B – Extended-response questions

Question 1 (11 marks) · state average 4.9 / 11 (45%)

2023 VCAA Exam 2 · Section B · Question 2

a. Show $b = \frac{\pi}{15}$ and $c = 75$

☐ Period 30: $\frac{2\pi}{b} = 30 \Rightarrow b = \frac{\pi}{15}$

☐ $h(0) = -60 + c = 15 \Rightarrow c = 75$

Answer / notes: Use the period and the lowest height.

State average 1.5 / 2 (75%); 66% scored full marks. Mark distribution: 0: 16% · 1: 18% · 2: 66%

Your mark: _____ / 2

b. Average height from A (bottom) to B (level with the centre) (2 d.p.)

☐ $\frac{1}{7.5} \int_0^{7.5} h(t) dt$

☐ $= 75 - \frac{120}{\pi} \approx 36.80$ m

Answer / notes: A quarter turn takes 7.5 minutes. ≈ 36.80 m

State average 0.9 / 2 (45%); 42% scored full marks. Mark distribution: 0: 54% · 1: 3% · 2: 42%

Your mark: _____ / 2

c. Average rate of change of height from A to B

☐ $\frac{75-15}{7.5} = 8$ m/min

Answer / notes: 8 m/min

State average 0.5 / 1 (50%); 55% scored full marks. Mark distribution: 0: 45% · 1: 55%

Your mark: _____ / 1

d.i. k and m in the piecewise model

☐ $k = 135, m = 2$

Answer / notes: It stops at the top (135 m) and restarts at double speed.

State average 0.4 / 1 (40%); 40% scored full marks. Mark distribution: 0: 60% · 1: 40%

Your mark: _____ / 1

d.ii. All possible values of n

☐ Continuity at $t = 20$: $h(40 + n) = 135$, so $\cos\left(\frac{\pi(40+n)}{15}\right) = -1$

☐ $n = 30j - 25$, $j \in \mathbb{Z}$

Answer / notes: $n = 5 + 30j$ for integer j . A general solution was required.

State average $0.4 / 2$ (20%); 12% scored full marks. Mark distribution: 0: 76% · 1: 12% · 2: 12%

Your mark: _____ / 2

d.iii. Sketch w with the coordinates of the endpoints

☐ Rising cosine from $(0, 15)$ to $(15, 135)$

☐ Flat from $(15, 135)$ to $(20, 135)$

☐ Half a (faster) cycle down to $(27.5, 15)$

Answer / notes: Three pieces joined continuously, with endpoints $(0, 15)$ and $(27.5, 15)$.

State average $1.2 / 3$ (40%); 24% scored full marks. Mark distribution: 0: 39% · 1: 25% · 2: 12% · 3: 24%

Your mark: _____ / 3

Question 2 (12 marks) · state average 5.3 / 12 (44%)

2023 VCAA Exam 2 · Section B · Question 3

a. $\lim_{x \rightarrow -\infty} g(x)$

☐ 5

Answer / notes: 5

State average $0.7 / 1$ (70%); 75% scored full marks. Mark distribution: 0: 25% · 1: 75%

Your mark: _____ / 1

b. k with $g'(x) = k \times 2^x$

☐ $k = \log_e 2$

Answer / notes: $k = \log_e 2$

State average $0.9 / 1$ (90%); 85% scored full marks. Mark distribution: 0: 15% · 1: 85%

Your mark: _____ / 1

c.i. Tangent at $(a, g(a))$

☐ $y = 2^a \log_e 2 (x - a) + 2^a + 5$

Answer / notes: Point–gradient form.

State average $0.5 / 1$ (50%); 52% scored full marks. Mark distribution: 0: 48% · 1: 52%

Your mark: _____ / 1

c.ii. Tangent to g through the origin (3 d.p.)

☐ Sets $0 = -a 2^a \log_e 2 + 2^a + 5$, giving $a \approx 2.618$

☐ $y \approx 4.255x$

Answer / notes: Substitute $(0, 0)$ into the general tangent, solve for a , then find the gradient.

State average $0.6 / 2$ (30%); 15% scored full marks. Mark distribution: 0: 52% · 1: 34% · 2: 15%

Your mark: _____ / 2

d. Point of inflection of h (2 d.p.)

☐ $\approx (2.06, -0.07)$

Answer / notes: $h''(x) = (\log_e 2)^2 2^x - 2 = 0$

State average $0.6 / 1$ (60%); 58% scored full marks. Mark distribution: 0: 42% · 1: 58%

Your mark: _____ / 1

e. Largest interval on which h is strictly decreasing (2 d.p.)

☐ $[0.49, 3.21]$

Answer / notes: Between the two roots of h' , with square brackets (the endpoints are included).

State average $0.4 / 1$ (40%); 35% scored full marks. Mark distribution: 0: 65% · 1: 35%

Your mark: _____ / 1

f. Newton's method from $x_0 = 0$: x_1, x_2, x_3 (3 d.p.)

☐ $x_1 \approx -1.443$

☐ $x_2 \approx -0.897, x_3 \approx -0.773$

Answer / notes: $-1.443, -0.897, -0.773$

State average 1.2 / 2 (60%); 54% scored full marks. Mark distribution: 0: 36% · 1: 10% · 2: 54%

Your mark: _____ / 2

g. Why a solution of $\log_e(2)^{2^x} - 2x = 0$ is a bad starting point

☐ That equation is $h'(x) = 0$, so the tangent is horizontal and Newton's formula divides by zero

Answer / notes: A stationary point makes the tangent horizontal, so it never meets the x-axis.

State average 0.2 / 1 (20%); 21% scored full marks. Mark distribution: 0: 79% · 1: 21%

Your mark: _____ / 1

h. Positive n for which $n^x - x^n$ has a local minimum on the x-axis

☐ Requires $f(x) = 0$ and $f'(x) = 0$ at the same x , giving $x = \frac{n}{\log_e n}$

☐ $n = e$

Answer / notes: $n = e$: the curve $e^x - x^e$ touches the axis at $x = e$. An exact answer was required.

State average 0.2 / 2 (10%); 3% scored full marks. Mark distribution: 0: 86% · 1: 12% · 2: 3%

Your mark: _____ / 2

Question 3 (11 marks) · state average 3.8 / 11 (35%)

2023 VCAA Exam 2 · Section B · Question 5

a. Complete the sequence of transformations from f to g

☐ Reflection in the y-axis

☐ Then translation 2 units right (or just translate 2 right, since f is even)

Answer / notes: Dilation $\frac{1}{2}$ from the x-axis, reflect in the y-axis, then translate right 2. Order matters.

State average 1.2 / 2 (60%); 35% scored full marks. Mark distribution: 0: 18% · 1: 48% · 2: 35%

Your mark: _____ / 2

b. Domain and range of the inverse of the increasing branch g_1

☐ Domain $[1, \infty)$

☐ Range $[2, \infty)$

Answer / notes: The domain and range of the inverse are swapped from g_1 (turning point $(2, 1)$).

State average 1.0 / 2 (50%); 39% scored full marks. Mark distribution: 0: 40% · 1: 21% · 2: 39%

Your mark: _____ / 2

c.i. Coordinates of P and Q (2 d.p.)

☐ $P \approx (1.27, 1.27), Q \approx (4.09, 4.09)$

Answer / notes: Solve $g(x) = x$. Both points lie on $y = x$.

State average 0.6 / 1 (60%); 55% scored full marks. Mark distribution: 0: 45% · 1: 55%

Your mark: _____ / 1

c.ii. Area bounded by g and the two inverses (2 d.p.)

☐ $2 \int_{1.27}^{4.09} (x - g(x)) dx$

☐ ≈ 5.56

Answer / notes: Use the symmetry about $y = x$: double the area between $y = x$ and g . Forgetting the 2 gives 2.78.

State average 0.6 / 2 (30%); 29% scored full marks. Mark distribution: 0: 65% · 1: 6% · 2: 29%

Your mark: _____ / 2

d. Integer n such that the turning point of h always lies on $y = 2x^n$

☐ $n = -1$

Answer / notes: The turning point is $(k, \frac{2}{k})$, and $\frac{2}{k} = 2k^{-1}$.

State average 0.1 / 1 (10%); 13% scored full marks. Mark distribution: 0: 87% · 1: 13%

Your mark: _____ / 1

e. Smallest k for which h meets the inverse of h_1 (2 d.p.)

☐ $k \approx 1.27$

Answer / notes: Find the k where the curves just touch (a tangency condition). The state averaged 0.0/1.

State average 0.0 / 1 (0%); 4% scored full marks. Mark distribution: 0: 96% · 1: 4%

Your mark: _____ / 1

f. Area between h and the inverse of h_1 when $k = 5$ (2 d.p.)

☐ Intersections at $x \approx 1.45$ and $x \approx 8.78$

☐ Area ≈ 43.91

Answer / notes: Integrate $|h - h_1^{-1}|$ between the two intersection points.

State average 0.3 / 2 (15%); 12% scored full marks. Mark distribution: 0: 86% · 1: 2% · 2: 12%

Your mark: _____ / 2

Question 4 (11 marks) · state average 4.9 / 11 (45%)

2024 VCAA Exam 2 · Section B · Question 5

a.i. Local maximum of $g \circ f$ in $[0, \frac{\pi}{2}]$ (1 d.p.)

☐ (0.9, 1)

Answer / notes: $\sin(2 \sin x) = 1$ when $\sin x = \frac{\pi}{4}$: $x \approx 0.9$.

State average 0.7 / 1 (70%); 69% scored full marks. Mark distribution: 0: 31% · 1: 69%

Your mark: _____ / 1

a.ii. Range of $g \circ f$ on $[0, 2\pi]$

☐ $[-1, 1]$

Answer / notes: $2 \sin x$ covers $[-2, 2]$, which contains $\pm \frac{\pi}{2}$, so the full $[-1, 1]$.

State average 0.6 / 1 (60%); 63% scored full marks. Mark distribution: 0: 37% · 1: 63%

Your mark: _____ / 1

b.i. Derivative of $f \circ g$

☐ $2 \cos(2x) \cos(\sin 2x)$

Answer / notes: Chain rule.

State average 0.8 / 1 (80%); 82% scored full marks. Mark distribution: 0: 18% · 1: 82%

Your mark: _____ / 1

b.ii. Show $\cos(\sin 2x) = 0$ has no real solutions

☐ It would need $\sin 2x = \frac{\pi}{2} + k\pi$

☐ But $|\sin 2x| \leq 1 < \frac{\pi}{2}$, so there are no solutions

Answer / notes: The sine can't reach $\pm \pi/2$.

State average 0.8 / 2 (40%); 34% scored full marks. Mark distribution: 0: 57% · 1: 9% · 2: 34%

Your mark: _____ / 2

b.iii. x-values of the stationary points of $f \circ g$ on $[0, 2\pi]$

☐ $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

Answer / notes: Only $\cos 2x = 0$ contributes (by part ii).

State average 0.5 / 1 (50%); 47% scored full marks. Mark distribution: 0: 53% · 1: 47%

Your mark: _____ / 1

b.iv. Range of $f \circ g$

☐ $[-\sin 1, \sin 1]$

Answer / notes: $\sin(\sin 2x)$ with $\sin 2x \in [-1, 1]$

State average 0.4 / 1 (40%); 34% scored full marks. Mark distribution: 0: 66% · 1: 34%

Your mark: _____ / 1

c.i. Single definite integral for the area between the two composites on $[0, 2\pi]$

☐ $\int_0^{2\pi} |\sin(\sin 2x) - \sin(2 \sin x)| dx$

Answer / notes: Use the absolute value so one integral covers every region.

State average 0.3 / 1 (30%); 32% scored full marks. Mark distribution: 0: 68% · 1: 32%

Your mark: _____ / 1

c.ii. That area (2 d.p.)

☐ ≈ 4.97

Answer / notes: ≈ 4.97

State average 0.5 / 1 (50%); 45% scored full marks. Mark distribution: 0: 55% · 1: 45%

Your mark: _____ / 1

d. x in $(0, 2\pi)$ for which $f_1 \circ g$ is defined, f_1 having domain $(0, 2\pi)$

☐ Requires $\sin 2x > 0$ (range of g inside the domain of f_1)

☐ $x \in (0, \frac{\pi}{2}) \cup (\pi, \frac{3\pi}{2})$

Answer / notes: The range of g must sit inside $(0, 2\pi)$, so $0 < \sin 2x$.

State average 0.3 / 2 (15%); 9% scored full marks. Mark distribution: 0: 80% · 1: 11% · 2: 9%

Your mark: _____ / 2

Question 5 (19 marks) · state average 8.5 / 19 (45%)

2025 VCAA Exam 2 · Section B · Question 4

a. $f(\frac{2\pi}{3})$

☐ $\frac{\sqrt{3}}{2} + 1$

Answer / notes: $\frac{\sqrt{3}}{2} + 1$

State average 0.9 / 1 (90%); 93% scored full marks. Mark distribution: 0: 7% · 1: 93%

Your mark: _____ / 1

b. Exact x with $f(x) = \frac{3}{2}$

☐ $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}$

Answer / notes: $\sin x = \frac{1}{2}$. Three solutions lie in the domain.

State average 0.8 / 1 (80%); 78% scored full marks. Mark distribution: 0: 22% · 1: 78%

Your mark: _____ / 1

c. k and the largest a with $f(x+k) = f(x)$ on $[0, a]$

☐ $k = 2\pi$

☐ Largest $a = \frac{\pi}{2}$

Answer / notes: The period is 2π , and $x + 2\pi$ must stay within $\frac{5\pi}{2}$.

State average 0.6 / 2 (30%); 15% scored full marks. Mark distribution: 0: 59% · 1: 26% · 2: 15%

Your mark: _____ / 2

d. Tangent at $x = \frac{2\pi}{3}$

☐ $y = -\frac{\pi}{2} + \frac{\pi}{3} + \frac{\sqrt{3}}{2} + 1$

Answer / notes: Gradient $\cos \frac{2\pi}{3} = -\frac{1}{2}$.

State average 0.7 / 1 (70%); 70% scored full marks. Mark distribution: 0: 30% · 1: 70%

Your mark: _____ / 1

e.i. Newton's method from $x_0 = \frac{2\pi}{3}$: x_2 to 1 d.p.

☐ $x_2 \approx 5.2$

Answer / notes: $x_1 = \frac{2\pi}{3} + \sqrt{3} + 2 \approx 5.83$, then $x_2 \approx 5.2$.

State average 0.6 / 1 (60%); 65% scored full marks. Mark distribution: 0: 35% · 1: 65%

Your mark: _____ / 1

e.ii. Draw the tangent at $x = x_1$ on the graph

- ☐ Tangent at $x_1 \approx 5.83$ drawn, meeting the x-axis at $x_2 \approx 5.2$

Answer / notes: The tangent at x_1 should cut the x-axis at about 5.2.

State average 0.3 / 1 (30%); 28% scored full marks. Mark distribution: 0: 72% · 1: 28%

Your mark: _____ / 1

f.i. Show the tangent at $x = p$ is $t(x) = \cos(p)(x - p) + \sin(p) + 1$

- ☐ Gradient $f'(p) = \cos p$ through $(p, \sin p + 1)$
☐ Correct point–gradient form rearranged

Answer / notes: $y - (\sin p + 1) = \cos p (x - p)$

State average 1.1 / 2 (55%); 49% scored full marks. Mark distribution: 0: 34% · 1: 17% · 2: 49%

Your mark: _____ / 2

f.ii. Minimum and maximum y-intercepts of t for $p \in (0, \frac{5\pi}{2})$

- ☐ y-intercept = $\sin p - p \cos p + 1$, derivative $p \sin p$, stationary at $\pi, 2\pi$
☐ Minimum $1 - 2\pi$, maximum $1 + \pi$

Answer / notes: At $p = 2\pi$: $1 - 2\pi$. At $p = \pi$: $1 + \pi$.

State average 0.5 / 2 (25%); 23% scored full marks. Mark distribution: 0: 72% · 1: 5% · 2: 23%

Your mark: _____ / 2

f.iii. p for which t has a unique x-intercept equal to f 's x-intercept (2 d.p.)

- ☐ Requires $t(\frac{3\pi}{2}) = 0$ with $p \neq \frac{3\pi}{2}$ (horizontal tangent excluded)
☐ $p \approx 2.38$ and $p \approx 7.04$

Answer / notes: Solve $\cos p (\frac{3\pi}{2} - p) + \sin p + 1 = 0$: $p \approx 2.38, 7.04$.

State average 0.4 / 2 (20%); 10% scored full marks. Mark distribution: 0: 65% · 1: 25% · 2: 10%

Your mark: _____ / 2

g.i. Show $c = 1$ and $d = 1$ given $g(0) = f(0)$, $g'(0) = f'(0)$

- ☐ $g(0) = d = f(0) = 1$
☐ $g'(0) = c = \cos 0 = 1$

Answer / notes: Direct substitution.

State average 1.4 / 2 (70%); 68% scored full marks. Mark distribution: 0: 27% · 1: 5% · 2: 68%

Your mark: _____ / 2

g.ii. Area between f and g on $[0, 2\pi]$ when they also match at 2π (2 d.p.)

- ☐ Solves for $a = \frac{1}{2\pi^2}$, $b = -\frac{3}{2\pi}$
☐ Area ≈ 1.53

Answer / notes: Use $\int_0^{2\pi} |f - g| dx \approx 1.53$.

State average 0.7 / 2 (35%); 29% scored full marks. Mark distribution: 0: 63% · 1: 8% · 2: 29%

Your mark: _____ / 2

g.iii. b and r with $g(r) = f(r)$, $g'(r) = f'(r)$ when $a = 0$, $c = d = 1$

- ☐ Sets up $br^2 + r = \sin r$ and $2br + 1 = \cos r$
☐ $r = \pi$, $b = -\frac{1}{\pi}$

Answer / notes: $r = \pi$ gives $b = -\frac{1}{\pi}$ (the other solution lies outside the domain).

State average 0.5 / 2 (25%); 14% scored full marks. Mark distribution: 0: 67% · 1: 19% · 2: 14%

Your mark: _____ / 2

Your result

Total: _____ / 126 State average on these questions: **49 / 126 (39%)**

Topics to revise:

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